

COMPUTER SCIENCE

PAPER 1 (THEORY)

Maximum Marks: 70

Time Allowed: Three Hours

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 12 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS

Answer all questions.

While answering questions in this Part, briefly indicate your working and reasoning, wherever required.

Question 1

- (i) The complement of the Boolean expression $P' \cdot (Q + R')$ is: (*Application*) [1]
- (a) $P + Q' \cdot R$
 - (b) $P' + Q' \cdot R$
 - (c) $P \cdot (Q' + R)$
 - (d) $P + Q \cdot R'$

- (ii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]

Assertion: A recursive method that has no base case ends with a `StackOverflowError` when it is called.

Reason: Every call that has not yet returned keeps its own activation record on the call stack, and the stack has limited memory.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (iii) According to the Principle of Duality, the Boolean equation $(P + 0) \cdot (P' + Q) = P \cdot Q$ will be equivalent to: (Application) [1]
(a) $(P \cdot 1) + (P' \cdot Q) = P + Q$
(b) $(P' \cdot 1) + (P \cdot Q') = P' + Q'$
(c) $(P + 1) \cdot (P' + Q) = P \cdot Q$
(d) $(P \cdot 0) + (P' \cdot Q) = P + Q$
- (iv) The law which states that $A + A \cdot B = A$ is: (Recall) [1]
(a) Idempotent law
(b) Absorption law
(c) Distributive law
(d) Complement law
- (v) The canonical form of the cardinal expression $F(A, B, C) = \pi(1, 4, 6)$ is: (Application) [1]
(a) $(A + B + C') \cdot (A' + B + C) \cdot (A' + B' + C)$
(b) $(A' + B' + C) \cdot (A + B' + C') \cdot (A + B + C')$
(c) $A'B'C + AB'C' + ABC'$
(d) $(A + B + C) \cdot (A' + B + C') \cdot (A' + B' + C')$
- (vi) Study the given propositions and the statements marked Assertion and Reason that follow them. Choose the correct option on the basis of your analysis. [1]

p = It rains heavily

q = The match is cancelled

s1 = $p \rightarrow q$

s2 = $\sim q \rightarrow \sim p$

Assertion: s2 is the contrapositive of s1.

Reason: A conditional statement and its contrapositive are logically equivalent.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.

- (vii) Consider the following statement written in class Visitor, where `count` is its data member. [1]

`static int count = 0;`

Which of the following statements are valid for `count`?

- I. It holds one value that is shared by all the objects of class Visitor.
- II. It can be used as `Visitor.count` even before any object is created.
- III. Every object of class Visitor gets its own separate copy of it.

(Application)

- (a) I and II
 - (b) II and III
 - (c) I and III
 - (d) Only III
- (viii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]

Assertion: After the statements `String s = "java"; s.toUpperCase();` the variable `s` still holds "java".

Reason: String objects are immutable; `toUpperCase()` returns a new String and leaves the original unchanged.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Both Assertion and Reason are false.
- (ix) State the time complexity, in Big O notation, of the following code segment, where `n` and `m` are the sizes of the inputs: [1]

```
for (int i = 0; i < n; i++)
    sum = sum + a[i];
for (int i = 0; i < m; i++)
    for (int j = 0; j < m; j++)
        c[i][j] = 0;
```

(Analysis)

- (x) Write the contrapositive of the statement: *If a number is divisible by 4, then it is even.* [1]
- (Application)

Question 2

- (i) Convert the following infix notation to prefix form. [2]
 $(A + B * C) / (D - E) * F$ (Create)
- (ii) A matrix $A[-3 \dots 12, 5 \dots 20]$ is stored in the memory with each element requiring 4 bytes of storage. If the base address is 3000, find the address of $A[7][14]$ when the matrix is stored in row major wise. (Application) [2]

- (iii) The following function `convert()` is a part of some class. Assume `n` is a positive integer. Answer the questions given below along with dry run / working.

```
int convert(int n)
{
    if (n == 0)
        return 0;
    return convert(n / 8) * 10 + n % 8;
}
```

- (a) What will the function `convert()` return when `n = 156`? (*Analysis*) [2]
(b) What is the function `convert()` performing apart from recursion? (*Analysis*) [1]
(iv) The following function `isNeon()` is a part of class `Neon`. It returns `true` if `n` is a Neon number, otherwise `false`. [A number is a Neon number if the sum of the digits of its square is equal to the number itself. Example: $9^2 = 81$ and $8 + 1 = 9$.]

There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a statement / expression so that the function works properly.

```
boolean isNeon(int n)
{
    int sq = ?1?;
    int sum = 0;
    while (sq > 0)
    {
        sum = sum + ?2?;
        sq = sq / 10;
    }
    return ?3?;
}
```

- (a) What is the expression or statement at ?1? (*Analysis*) [1]
(b) What is the expression or statement at ?2? (*Analysis*) [1]
(c) What is the expression or statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) A cricket academy selects players for its under-19 squad. A player is selected if any one of the following conditions is satisfied: [5]

- The player has cleared the fitness test and has a batting average above 40, but has taken fewer than 20 wickets in the season.

OR

- The player has cleared the fitness test, has taken 20 or more wickets and is recommended by the coach.

OR

- The player has not cleared the fitness test, but has a batting average above 40, has taken 20 or more wickets and is recommended by the coach.

OR

- The player has cleared the fitness test and is recommended by the coach, but neither has a batting average above 40 nor has taken 20 or more wickets.

The inputs are:

INPUTS	
C	has cleared the fitness test
B	has a batting average above 40
W	has taken 20 or more wickets in the season
R	is recommended by the coach

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: X – denotes the player is selected [1 indicates YES and 0 indicates NO in all cases]

Draw the truth table for the inputs and outputs given above and write the SOP expression for $X(C, B, W, R)$.

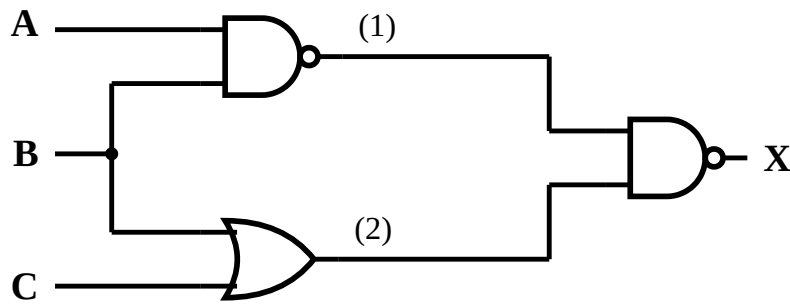
(Application)

- (ii) Reduce the above expression $X(C, B, W, R)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. (Create) [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(P, Q, R, S) = \pi(0, 1, 2, 3, 4, 6, 8, 12, 14)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). (Application) [4]
- (b) Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. (Create) [1]

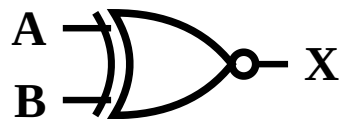
- (ii) From the logic diagram given below:



- (a) Derive the Boolean expression for the output X, reduce it, and draw the truth table for the derived expression. (*Application*) [4]
- (b) If $A = 1$, $B = 0$ and $C = 1$, then find the value of X. (*Application*) [1]

Question 5

- (i) Draw the logic circuit to decode the following binary numbers (0010, 0110, 1001, 1011, 1101, 1111) to their hexadecimal equivalents. Also state the hexadecimal equivalents of the given binary numbers. (*Application & Analyse*) [5]
- (ii) Verify if the following proposition is valid using the truth table:
 $(P \vee Q) \Rightarrow R = (P \Rightarrow R) \wedge (Q \Rightarrow R)$ (*Application*) [3]
- (iii) Answer the following questions related to the gate given below:



- (a) What is the output of the above gate if the inputs are $A = 1$, $B = 0$? (*Analysis*) [1]
- (b) What are the values of the inputs if the output is 1? (*Application*) [1]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem. This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

A **Disarium number** is a number in which the sum of its digits, each raised to the power of its position (counted from the left, starting at 1), is equal to the number itself. [10]

Example: $135 = 1^1 + 3^2 + 5^3 = 1 + 9 + 125 = 135$

$175 = 1^1 + 7^2 + 5^3 = 1 + 49 + 125 = 175$

Design a class **Disarium** to check if a given number is a Disarium number or not. Some of the members of the class are given below:

Class name : Disarium

Data members/instance variables:

num : to store the number

size : to store the number of digits of num

Methods / Member functions:

Disarium(int nn) : parameterised constructor to initialise num = nn and size = 0

void countDigit() : counts the number of digits of num and stores it in size

int sumofDigits(int n, int p) : returns the sum of the digits of n, each raised to the power of its position, using the **recursive technique**, where p is the position of the last digit of n

void check() : checks whether num is a Disarium number by invoking sumofDigits() and displays an appropriate message

Specify the class Disarium giving details of the constructor(int), void countDigit(), int sumofDigits(int, int) and void check(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 7

A class **Merge** contains an array of integers arranged in ascending order. Two such arrays are to be merged into a third array which is also in ascending order. The merging must be done by comparing the elements of the two arrays, *without* sorting the resultant array afterwards. [10]

Example: First array: { 3, 8, 15, 42 }

Second array: { 1, 8, 20 }

Merged array: { 1, 3, 8, 8, 15, 20, 42 }

The details of the members of the class are given below:

Class name : Merge

Data members/instance variables:

arr[] : integer array

size : integer to store the number of elements

Member functions/methods:

Merge(int s) : parameterised constructor to assign size = s and create the array

void input() : to accept the elements of the array in ascending order

Merge join(Merge P, Merge Q) : returns an object whose array contains all the elements of the arrays of objects P and Q in ascending order, formed by merging them

void display() : to display the array

Specify the class Merge giving details of the constructor(int), void input(), Merge join(Merge, Merge) and void display(). Define a main() function to create objects and call all the functions accordingly to enable the task.

(Create)

Question 8

A class **SortLen** has been defined to arrange the words of a sentence in ascending order of their lengths. Words of the same length must remain in the order in which they occur in the sentence. The sentence is in upper case, its words are separated by a single blank space and it is terminated by '.', '?' or '!'. If the sentence is not terminated by one of these, display "INVALID INPUT".

[10]

Example 1: INPUT: LEARN JAVA TO BUILD NEW APPS.

OUTPUT: TO NEW JAVA APPS LEARN BUILD.

Example 2: INPUT: HELLO WORLD

OUTPUT: INVALID INPUT

Some of the members of the class are given below:

Class name : SortLen

Data members/instance variables:

sent : to store the sentence

newsent : to store the rearranged sentence

Methods / Member functions:

SortLen() : default constructor to initialise the data members with legal initial values

void accept() : to accept the sentence in upper case

boolean isValid() : returns true if the sentence is terminated by '.', '?' or '!', otherwise returns false

void arrange() : arranges the words of sent in ascending order of their lengths, keeping words of equal length in their original order, and stores the result, with the same terminating character, in newsent

void display() : displays the original and the rearranged sentence

Specify the class SortLen giving details of the constructor(), void accept(), boolean isValid(), void arrange() and void display(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise. This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms.

The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9

A single-lane **Tunnel** on a hill road allows vehicles to enter or leave from either of its two ends. It can hold at most a fixed number of vehicles, each identified by an integer registration number.

Define a class Tunnel with the following details:

Class name : Tunnel

Data members/instance variables:

car[] : array to hold the registration numbers of the vehicles

cap : stores the maximum capacity of the array

front : to point to the index of the vehicle at the front end

rear : to point to the index of the vehicle at the rear end

Methods / Member functions:

Tunnel(int max) : constructor to initialise cap = max, front = rear = -1 and to create the integer array

void addFront(int v) : to add a vehicle at the front end if possible, otherwise display the message "NO ENTRY FROM FRONT"

int removeFront() : to remove and return the vehicle at the front end; if the tunnel is empty, return -999

void addRear(int v) : to add a vehicle at the rear end if possible, otherwise display the message "NO ENTRY FROM REAR"

int removeRear() : to remove and return the vehicle at the rear end; if the tunnel is empty, return -999

- (i) Specify the class Tunnel giving details of the functions void addRear(int) and int removeFront(). Assume that the other functions have been defined. The main() function and algorithm need NOT be written. *(Create)* [4]
- (ii) Name the entity described above and state its principle. *(Understanding)* [1]

Question 10

A multi-level car park charges vehicles according to the number of hours parked, as shown below:

[5]

Hours parked	Charge (Rs)
Up to 2 hours	50 (flat)
More than 2 and up to 5 hours	50 + 30 for each hour beyond 2
More than 5 hours	140 + 40 for each hour beyond 5

A vehicle of type "SUV" is charged 20% more than the amount above.

A super class **Vehicle** has been defined to store the details of a vehicle. Define a sub class **Parking** to calculate the parking charge. The details of the members of both the classes are given below:

Class name : Vehicle

Data members/instance variables:

vno : to store the registration number of the vehicle

type : to store the type of the vehicle ("CAR" or "SUV")

hrs : to store the number of hours parked (integer)

Methods / Member functions:

Vehicle(...) : parameterised constructor to assign values to the data members

void show() : to display the vehicle details

Class name : Parking

Data members/instance variables:

charge : to store the parking charge (in decimals)

Methods / Member functions:

Parking(...) : parameterised constructor to assign values to the data members of both the classes

void compute() : calculates the parking charge as per the table and the rule given above

void show() : displays the vehicle details along with the parking charge

Assume that the super class Vehicle has been defined. Using the concept of inheritance, specify the class Parking giving the details of the constructor(...), void compute() and void show().

The super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Student. The class structure of the Student is given below: [2]

```
class Student
{
    String name;
    int marks;
    Student next;
}
```

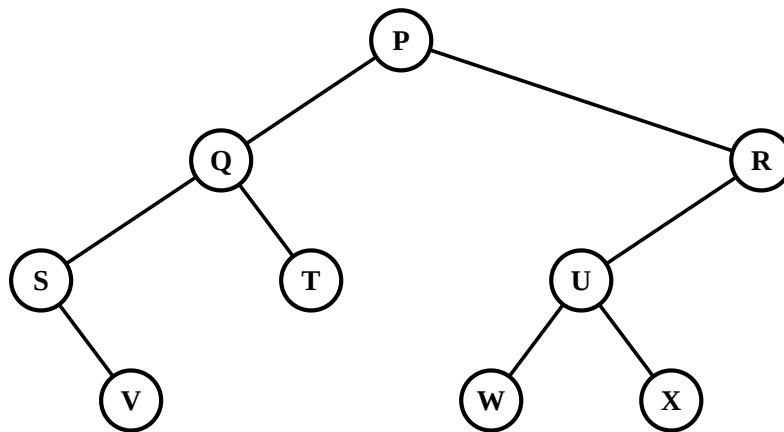
Write an Algorithm OR a Method to count and display the number of students who have scored more than 90 marks.

The method declaration is as follows:

```
void countTop(Student first)
```

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Tree given below:



- (a) Write the post-order traversal of the above tree structure. (Understanding) [1]
(b) State the depth of the tree and the level of node W, when the root is at level 0. (Understanding) [1]
(c) Name the external nodes of the right subtree and state the degree of node U. (Understanding) [1]

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